BDA Project Report

Table of Contents

BDA PROJECT REPORT **1**

1. Introduction **2**

2. Description of the data and the analysis problem. **2**

3. Description of two models **5**
3.1. Non-hierarchical model 5
3.2. Hierarchical model 5

4. The weakly informative priors used and the justification of these priors **6**
4.1. Non-hierarchical model 6
4.2. Hierarchical model 7

5. Brms code **8**
5.1. Non-hierarchical model 8
5.2. Hierarchical model 8

6. How the MCMC inference was run **9**

7. Convergence diagnostics **9**
7.1. Non-hierarchical model 9
7.2. Hierarchical model 10

8. Posterior predictive checks **11**
8.1. Posterior predictive checks for non-hierarchical model 11
8.2. Posterior predictive checks for hierarchical model 13

9. Predictive performance assessment **15**
9.1. Non-hierarchical model 15
9.2. Hierarchical model 16

10. Sensitivity analysis **16**
10.1. Non-hierarchical model 16
10.2. Hierarchical model 18

11. Model comparison **18**

12. Discussion of issues and potential improvements **19**

13. Conclusion **20**

14. Self-reflection **20**

15. References **21**

1. Introduction

The motivation behind this report is to understand server pointwise win rates in tennis matches. The point system in tennis is illustrated in Figure 1 shows how a match consists of sets, which consists of games, which in turn consist of individual points. Certain points in a tennis match are, however, more important than others in deciding the match. The problem in this report is to specifically understand how pressure affects server pointwise win rates by looking at these more critical points. This is measured by looking at break points and tiebreak points. A breakpoint is a situation where the returner of the serve has a chance to win the game by winning the next point. Because the serving player is usually favoured to win the game, facing a breakpoint could be expected to create pressure, thus potentially reducing the probability of winning the game. A tiebreak on the other hand is used to decide a set when the score in games has reached 6-6. The first player to reach seven points or more and win by two points wins the tiebreak and therefore the set. As three sets are required to win the tennis match, each tiebreak point becomes critical and can therefore be assumed to create more pressure for the player.

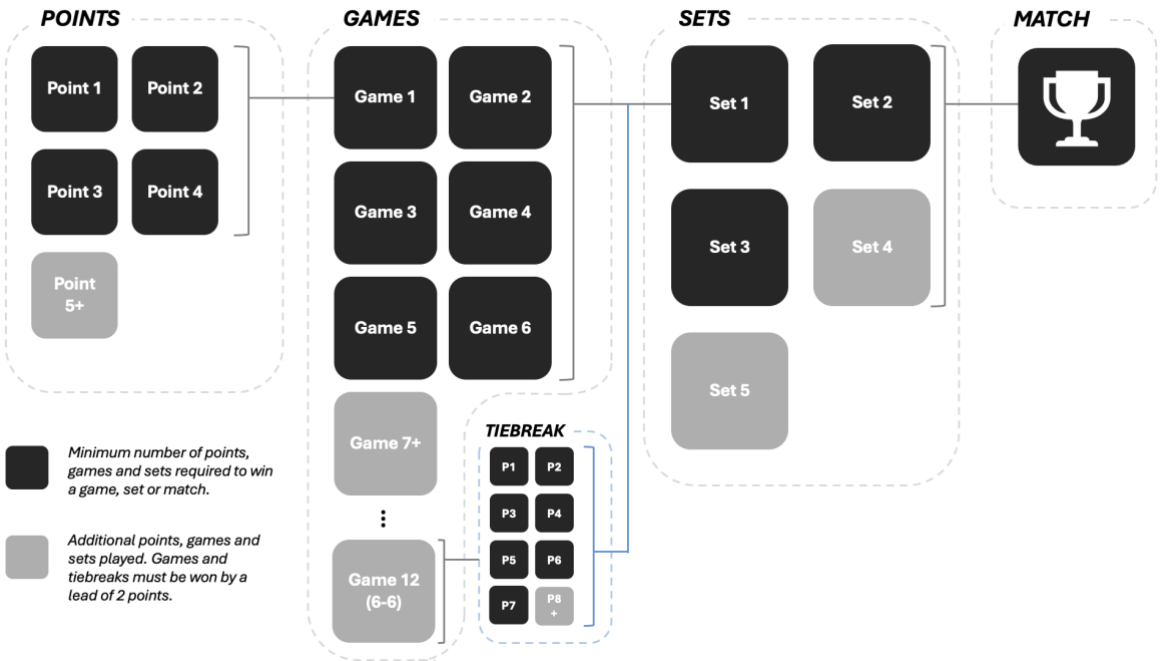

Figure 1: Point system in a tennis match.

The main modelling idea, given the abovementioned problem definition, is to use a binomial model to estimate the unknown proportion of a population from data results $y_1, \dots, y_n$, which take the value either 0 or 1 (Gelman et al., 2021). Here 1 indicates the server won the point and 0 indicates the server lost the point. The parameter π is defined as the ratio of points won and lost by the server under pressure. We assume that the n recorded points in the dataset used are conditionally independent given π, as we e.g. don't have any data which could explain how previous points could affect the pressure and thus the probability of winning a specific point for the server.

2. Description of the data and the analysis problem.

We are using Jeff Sackmann's Tennis Slam Point-by-Point data set, which consists of point-by-point account of tennis matches from four major tournaments dating back to 2011 (Sackmann, n.d.). Due to the vast amount of data in the set, we had to choose a subset. For

our own interest, we selected 10 of the highest rated players, and chose a random subset of matches such that, in the resulting dataset, each of our selected players is guaranteed to play 12 matches. It should be noted that, while we designed the dataset to contain matches with these 10 players, the performance of *every* player (read: selected players and their opponents) was still included in the model, with the understanding that player-specific information about the non-selected players will be highly unreliable. The justification for this is that we are analysing the effect of pressure on win probability, and the data of every player still gives valuable population level information.

The resulting sample contains 104 full matches, with 76 distinct players, consisting of a total of 24 267 *points played*, which in our analysis represent single observations. Figure 2 shows the total number of points played in the dataset, along with the proportion of points played under pressure, defined as breakpoints and points played on tiebreak rounds. Figure 3 shows the number of matches played at each tournament, as well as the distribution of number of games played by a specific player. We were not able to find any existing analyses similar to ours.

A preliminary analysis of the win probability at regular points, breakpoints, and tiebreak points, averaged over entire matches, yielded the distributions shown in Figure 4 and Figure 5, which indicates that they differ a fair amount. Due to the small relative number of breakpoint and tiebreak observations compared to the full dataset, we expect the variance of those win probability distributions to be higher than that of the win probability of regular points, which we indeed see in the figures. However, at least in the case of breakpoint win probability, the distributions seem to differ in skewness and mean as well, giving a justification to further study this phenomenon using Bayesian methods.

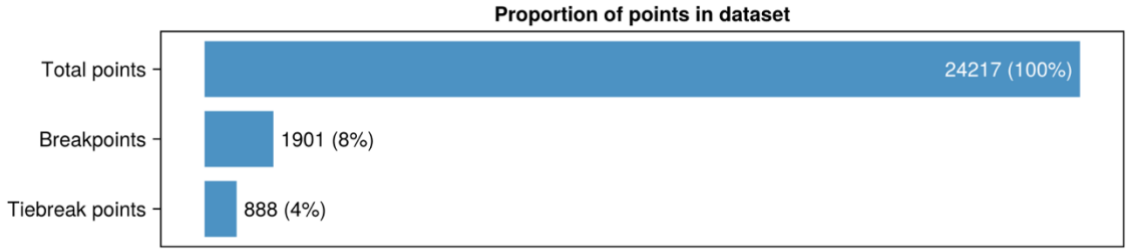

Figure 2: Proportion of points played in the dataset. Breakpoints and tiebreak points comprise a relatively small fraction of the total points played.

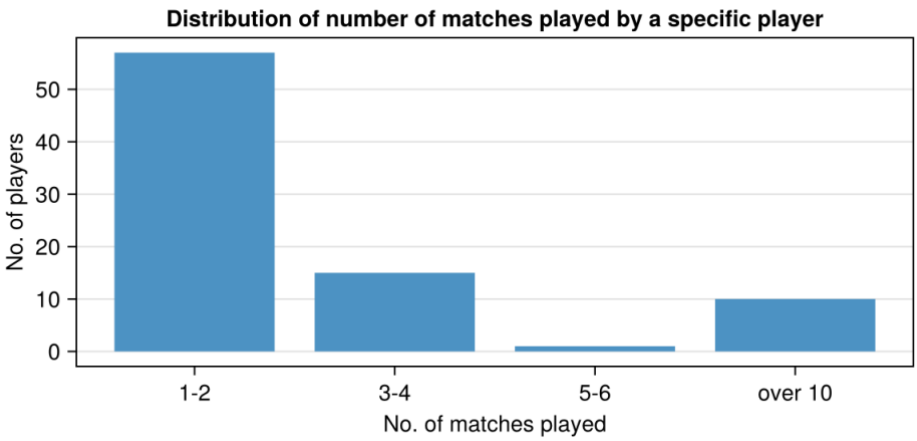

Figure 3: Distribution of number of matches played by a specific player. Most players in our dataset appear in only one or two matches.

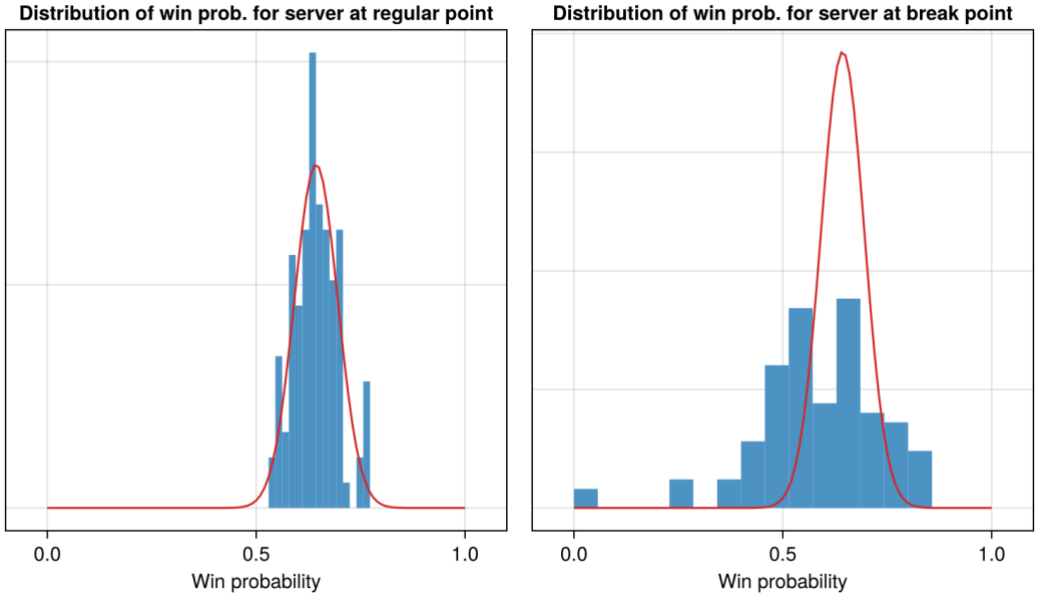

Figure 4: Distribution of win probability for serving player on any point (left) versus on a break point (right). The red curve is a normal approximation of win probability for the server at a regular point.

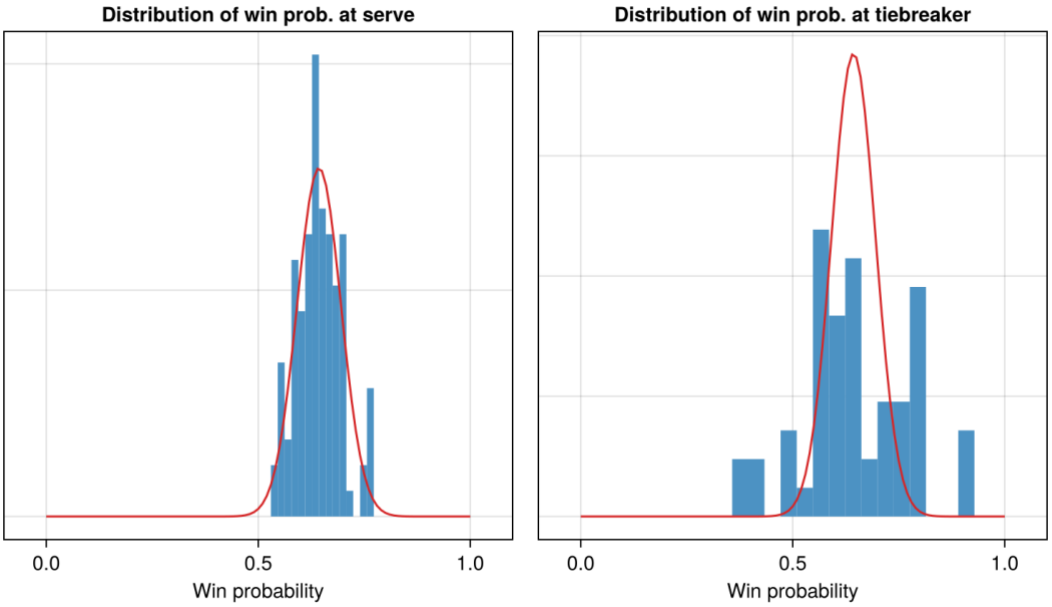

Figure 5: Distribution of win probability for serving player on any point (left) versus on a point in a tiebreaker game (right). The red curve is a normal approximation of win probability for the server at a regular point.

3. Description of two models

3.1. Non-hierarchical model

To assess how pressure affects server pointwise win rates, a binomial prediction model is defined as

$$y_i|\pi_i \sim \mathrm{Ber}(\pi_i) \quad (1),$$

for $i = 1, \ldots, n$ data points. Then,

$$\mathrm{logit}(\pi_i) = \alpha + \beta_{\mathrm{bp}} is_bp_i + \beta_{\mathrm{tb}} is_tb_i \quad (2),$$

where α is the baseline log-probability that the server wins any point, not necessarily a break point or tiebreak point. β_{bp} shifts the log-probability on break points against the server, while β_{tb} shifts the log-probability for the server on tiebreak points. For simplicity, is_bp_i takes the value 1 on break points against the server, and 0 otherwise. Similarly, is_tb_i takes the value 1 on tiebreak points for the server, and 0 otherwise. These two are mutually exclusive, as there are no break points during tiebreaks. The likelihood function becomes

$$p(y|\alpha, \beta_{\mathrm{bp}}, \beta_{\mathrm{tb}}) = \prod_{i=1}^{n} \mathrm{Ber}(y_i|\mathrm{logit}^{-1}(\alpha + \beta_{\mathrm{bp}} is_bp_i + \beta_{\mathrm{tb}} is_tb_i)) \quad (3),$$

while the posterior is then formulated as

$$p(\alpha, \beta_{\mathrm{bp}}, \beta_{\mathrm{tb}}|y) \propto p(\alpha)p(\beta_{\mathrm{bp}})p(\beta_{\mathrm{tb}}) \prod_{i=1}^{n} \mathrm{Ber}(y_i|\mathrm{logit}^{-1}(\alpha + \beta_{\mathrm{bp}} is_bp_i + \beta_{\mathrm{tb}} is_tb_i)) \quad (4),$$

where $p(\alpha), p(\beta_{\mathrm{bp}}), p(\beta_{\mathrm{tb}})$ are the priors for the covariates. They are discussed in section 4.

3.2. Hierarchical model

A hierarchical model is further created based on differences between players to expand on the first model. First, it is assumed that each player has their own serve winning probability for a normal point without pressure. Thus, the intercept will be varied for each player. Furthermore, we assume that each player handles pressure differently, when serving in a break or tiebreak point. Thus, the slopes for is_bp_i and is_tb_i will varied for each player. The hierarchical model can, based on this, be defined as:

$$y_i|\pi_i \sim \mathrm{Ber}(\pi_i) \quad (5),$$

for $i = 1, \ldots, n$ data points. Then for $j = 1, \ldots, m$ players,

$$\mathrm{logit}(\pi_i) = \alpha + u_{0,i}^{(j)} + \left(\beta_{\mathrm{bp}} + u_{\mathrm{bp},i}^{(j)}\right) is_bp_i + \left(\beta_{\mathrm{tb}} + u_{\mathrm{tb},i}^{(j)}\right) is_tb_i \quad (6),$$

where α is again the baseline log-probability that the server wins any point, β_{bp} is on a population-level and shifts the log-probability on break points against the server, and β_{tb} is similarly on a population-level and shifts the log-probability for the server on tiebreak points. Furthermore, $u_{0,i}^{(j)}$ is the server-specific intercept, $u_{\mathrm{bp},i}^{(j)}$ the server-specific deviation in the

slope in break points and $u_{\mathrm{tb,i}}^{(\mathrm{j})}$ the server-specific deviation in the slope in tiebreak points. The likelihood function becomes

$$p(y|\alpha, \beta_{\mathrm{bp}}, \beta_{\mathrm{tb}}, \mathbf{u}^{(j)}) = \prod_{i=1}^{n} \mathrm{Ber}(y_i|\mathrm{logit}^{-1}(\eta_i)), \tag{7}$$

where

$$\eta_{\mathrm{i}} = \alpha + u_{0,i}^{(j)} + (\beta_{\mathrm{bp}} + u_{\mathrm{bp},i}^{(j)}) is_bp_i + (\beta_{\mathrm{tb}} + u_{\mathrm{tb},i}^{(j)})\, is_tb_i, \tag{8}$$

$$\mathbf{u}^{(j)} = \begin{pmatrix} u_0^{(j)} \\ u_{\mathrm{bp}}^{(j)} \\ u_{\mathrm{tb}}^{(j)} \end{pmatrix}, \tag{9}$$

and the posterior is defined as

$$p\left(\alpha, \beta_{\mathrm{bp}}, \beta_{\mathrm{tb}}, \{\mathbf{u}^{(j)}\}, \Sigma \mid y\right)$$
$$\propto\ p(\alpha)p(\beta_{\mathrm{bp}})p(\beta_{\mathrm{tb}})p(\Sigma)p\left(y|\alpha, \beta_{\mathrm{bp}}, \beta_{\mathrm{tb}}, \{\mathbf{u}^{(j)}\}\right) \prod_{j=1}^{m} p\left(\mathbf{u}^{(j)} \mid \Sigma\right), \tag{10}$$

where Σ is the 3x3 covariance matrix and $p(\alpha)$, $p(\beta_{bp})$, $p(\beta_{tb})$, $p(\Sigma)$, $\mathrm{p}\left(\mathbf{u}^{(j)} \mid \Sigma\right)$, are the priors for the covariates. The priors for $\mathbf{u}^{(j)}$ includes hyperparameters, which for notation simplicity are not included in the posterior above, but instead defined through the priors of $\mathbf{u}^{(j)}$ in Section 4.2. In addition to this, the priors are further outlined in detail in Section 4.

4. The weakly informative priors used and the justification of these priors

4.1. Non-hierarchical model

For the non-hierarchical model in defined in section 3.1 and equations (1)-(4), we use weakly informative normal priors on with logit scale for the regression parameters. Specifically, we set

$$p(\alpha) \sim \mathcal{N}(0, 2^2), \qquad p(\beta_{\mathrm{bp}}), p(\beta_{\mathrm{tb}}) \sim \mathcal{N}(0, 1^2), \tag{11}$$

independently. The intercept prior for α implies that the baseline serve win probability is likely to be roughly between 0.1 and 0.9, which is deliberately vague yet ruling out extreme values close to 0 or 1 that would be unrealistic. The priors on β_{bp} and β_{tb} represent ratios roughly between 1/8 and 8, allowing for substantial effects of break points and tiebreaks without assuming unrealistically large impacts a priori. For simplicity, variances are assumed to be fixed for the priors.

From BDA3, weakly informative normal priors for logistic regression coefficients provide regularization, improve computational stability, and prevent the posterior from concentrating

on unrealistic regions, while still allowing the likelihood to dominate given our relatively large dataset (Gelman et al., 2021).

4.2. Hierarchical model

For the hierarchical model the weakly informative priors outlined in equation (11) are used. Weakly informative priors are also used for the player-specific random effects. We set these prior for the server specific random effects as

$$p\left(\mathbf{u}^{(j)} \mid \Sigma\right) \sim \mathcal{N}_3(0, \Sigma), \tag{12}$$

where

$$u_0^{(j)} \sim \mathcal{N}(0, \tau_0^2), \qquad u_{\mathrm{bp}}^{(j)} \sim \mathcal{N}\left(0, \tau_{\mathrm{bp}}^2\right), \qquad u_{\mathrm{tb}}^{(j)} \sim \mathcal{N}\left(0, \tau_{\mathrm{tb}}^2\right), \tag{13}$$

and

$$p(\tau_0),\ p(\tau_{\mathrm{bp}}),\ p(\tau_{\mathrm{tb}}) \sim t_3^+(0, 2.5), \tag{14}$$

while the prior for the covariance matrix is set as

$$p(\Sigma) = p(\tau_0) p\left(\tau_{\mathrm{bp}}\right) p(\tau_{\mathrm{tb}}) p(R), \tag{15}$$

where

$$R \sim LKJ(1), \tag{16}$$

We assume the player specific differences $\mathbf{u}^{(j)}$ follows a multivariate normal distribution and we centre this around 0 to not assume any deviation across all players. The priors for the variance between players in terms of serving and pressure handling, $p(\tau_0)$, $p(\tau_{\mathrm{bp}})$ and $p(\tau_{\mathrm{tb}})$, are set as a Student-t+ distribution, denoted as t_ν^+, as this is the default prior in brms for random effect standard deviation and this weakly informative prior enables a wide range of values. For the covariance matrix we also do not assume any direct correlations between the performance in regular, break point and tiebreak point serves from the start and thus again use the brms default prior for correlations $LKJ(1)$.

5. Brms code

5.1. Non-hierarchical model

The model was implemented using brms. Following data cleaning, the first model is implemented as below.

```
fit <- brm(
    formula = ServerWon ~ BreakPoint + Tiebreaker,
    data = df,
    family = bernoulli(link = "logit"),
    prior = c(
      prior(normal(0,2^2), class = "Intercept"),
      prior(normal(0,1^2), class = "b")
    ),
    chains = 4,
    iter = 2000,
    cores = 4
)
```

Figure 6: brms implementation of the non-hierarchical model.

5.2. Hierarchical model

The hierarchical model was further also implemented using brms. The implementation of this second model can be seen below.

```
fit_hier <- brm(
  formula = ServerWon ~ BreakPoint + Tiebreaker +
            (1 + BreakPoint + Tiebreaker | ServerID),
  data    = df,
  family  = bernoulli(link = "logit"),
  prior   = c(
    # fixed effects (α, β_bp, β_tb)
    prior(normal(0, 2^2), class = "Intercept"),
    prior(normal(0, 1^2), class = "b"),

    # random-effect standard deviations and correlation
    # for ServerID - brms defaults
    prior(student_t(3, 0, 2.5), class = "sd", group = "ServerID"),
    prior(lkj(1), class = "cor", group = "ServerID")
  ),
  chains  = 4,
  cores   = 4,
  iter    = 2000
)
```

Figure 7: brms implementation of the hierarchical model.

6. How the MCMC inference was run

As seen in Figure 6 and Figure 7, standard brms options were used when running the MCMC inference. Thus, we used 4 chains with 2000 iterations per chains, resulting in 4000 draws in total. We believe this was sufficient for stable estimates of our models. We deliberately opted to not choose a higher number of chains or iterations, as our dataset was quite large (approx. 24 000 data points), and a higher number of iterations would have added unnecessary computational strain. All other sampler options, such as step size adaptation and maximum tree depth, were left as default brms values.

7. Convergence diagnostics

7.1. Non-hierarchical model

Diagnostics for the fit of the first model can be seen below in Table 1.

Parameter	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.58	0.01	0.56	0.61	1.00	5041	2929
BreakPoint	-0.14	0.05	-0.23	-0.04	1.00	4117	3111
Tiebreaker	0.10	0.08	-0.05	0.26	1.00	4756	3182

Table 1: convergence diagnostics of the non-hierarchical model

The convergence diagnostics in Table 1 indicate that the model has sampled the posterior distribution reliably. The $\hat{R}$ value for each of the three parameters is 1.00, which implies that the Markov chains have mixed well and are sampling from the same stationary distribution. Moreover, the ESS, both bulk and tail, for each parameter, are high. The recommended minimum, according to (Vehtari et al., 2021), for four chains is 400, and our ESS are all comfortably over 2500, which exceeds threshold. Thus, our $\hat{R}$ values are likely to be stable.

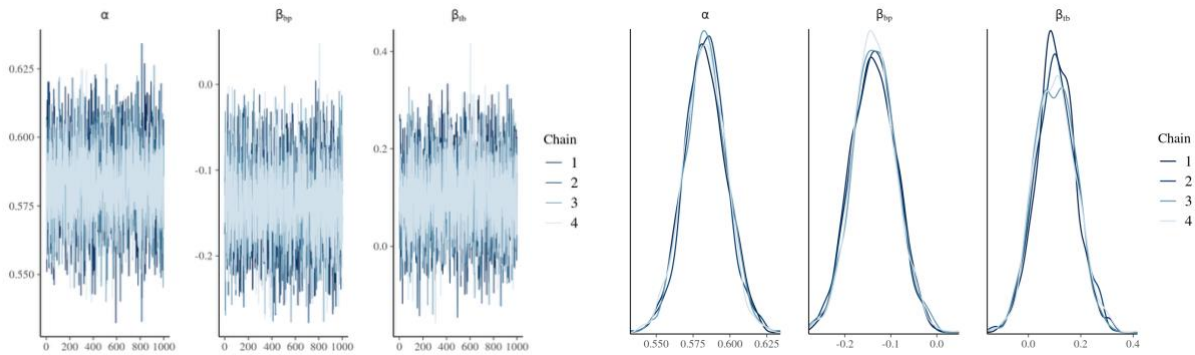

Figure 8: MCMC trace and density overlay plot for the parameters of the non-hierarchical model.

The reported posterior means and 95% credible intervals can hence be interpreted with confidence. For instance, the BreakPoint parameter's credible interval is below zero, indicating that it has a negative effect on server win probability. On the other hand, the Tiebreaker parameter's credible interval ranges from -0.05 to 0.26, indicating that serve pressure effects during tiebreaks may not have a large impact on serve point win probabilities. Additionally, no divergences or max tree depth hits were observed. Thus, we

can be confident that the model has converged satisfactorily. The plots in Figure 8 also show stable convergence behaviour.

7.2. Hierarchical model

Diagnostics for the fit of the second model can be seen below in Table 2. Table 2 outlines the diagnostics for the server-specific random effects as well as the fixed population-level effects.

Parameter	**Estimate**	**Est. Error**	**l-95% CI**	**u-95% CI**	**Rhat**	**Bulk_ESS**	**Tail_ESS**
sd(Intercept)	0.20	0.03	0.14	0.27	1.00	1291	2462
sd(BreakPoint)	0.09	0.07	0.00	0.24	1.00	1296	2058
sd(Tiebreaker)	0.36	0.14	0.08	0.64	1.00	1266	1175
cor(Intercept, BreakPoint)	0.15	0.46	-0.80	0.90	1.00	3948	2691
cor(Intercept, Tiebreaker)	-0.09	0.37	-0.75	0.64	1.00	3249	2713
cor(BreakPoint, Tiebreaker)	-0.14	0.47	-0.91	0.79	1.01	1005	1801
Intercept	0.49	0.03	0.43	0.56	1.00	1463	2308
BreakPoint	-0.12	0.06	-0.23	-0.02	1.00	4042	2777
Tiebreaker	0.09	0.12	-0.15	0.33	1.00	3602	2971

Table 2: Convergence diagnostics for the hierarchical model, server-level random effects and population-level fixed effects.

The convergence diagnostics in Table 2 indicate that the model has sampled the posterior distribution reliably. The $\hat{R}$ value for each of the three parameters is 1.00-1.01, which implies that the Markov chains have mixed well. Additionally, the ESS bulk and tail parameters exceed the recommended minimum of 400 given the use of four chains (Vehtari et al., 2021). This indicates that the posterior distribution is explored efficiently and can be trusted in terms of the tails, means and quantiles. Furthermore, no divergences or max tree depth hits were observed. This stable behaviour can further be seen in Figure 9 and Figure 10.

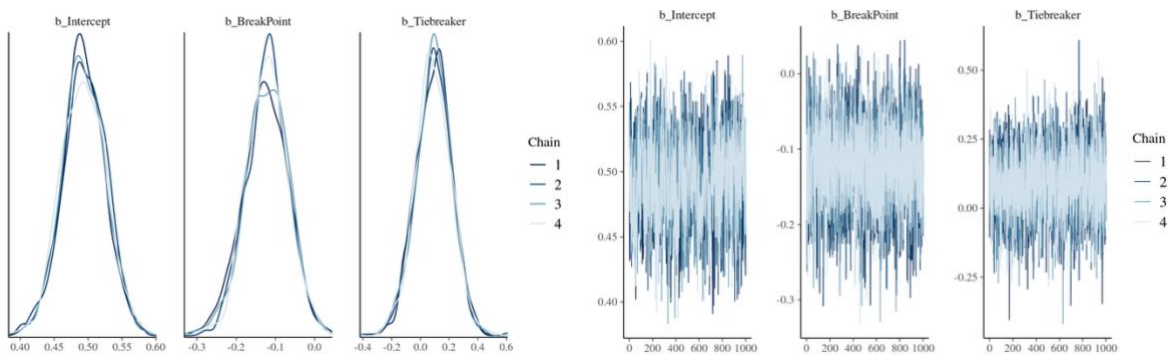

Figure 9: MCMC trace and overlay plots for the population parameters of the hierarchical model.

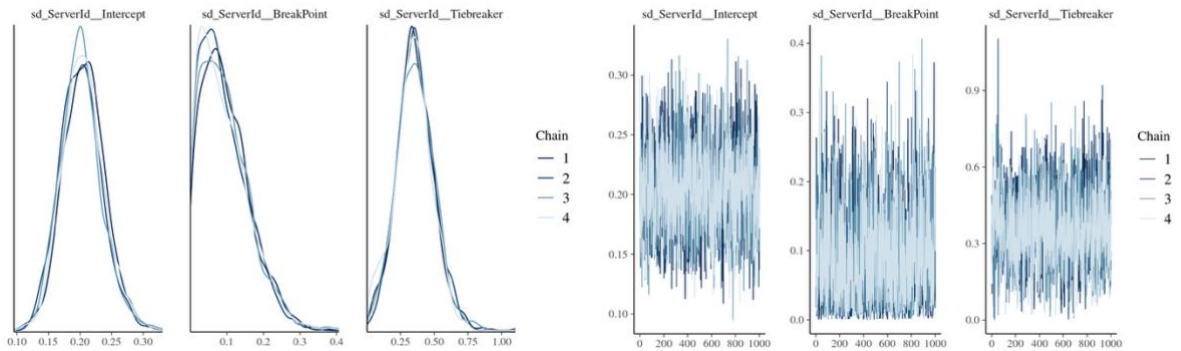

Figure 10: MCMC trace and overlay plots for the population parameters of the hierarchical model.

Furthermore, when comparing diagnostics of the non-hierarchical model in Table 1 with the diagnostics of the hierarchical model in Table 2 the following can be concluded. The baseline win probability determined by the intercept varies slightly: 0.58 (0.01) for the first model and 0.49 (0.03) for the second. The effect of the break points, −0.14 (0.05) vs. −0.12 (0.06), and the tiebreak points, 0.10 (0.08) vs. 0.09 (0.12), are, however, quite similar.

8. Posterior predictive checks

8.1. Posterior predictive checks for non-hierarchical model

To evaluate the fit of the non-hierarchical model, we chose to use PSIS-LOO metrics, reliability diagrams, LOO-PIT diagnostics and posterior predictive distribution visualizations.

As seen in Table 3, the PSIS-LOO for the first model is -15855.4, with an effective number of parameters of 3.0, reflecting the nominal number of regression coefficients. The looic of 31710.8 implies that the model is reasonably greedy and does not appear to overfit the data.

Measure	**Estimate**	**SE**
elpd_loo	-15855.4	43.1
p_loo	3.0	0.0
looic	31710.8	86.3

Table 3: PSIS-Loo diagnostics of the non-hierarchical model.

The reliability diagram in Figure 11 compares binned predicted win probabilities to the observed frequencies. The fitted curve and the points lie close to the 45-degree line, which suggests that the predicted probabilities are well calibrated. The LOO-PIT-ECDF in Figure 12 follows the reference line closely, with some minor deviations. This indicates that the predictive distribution is consistent with the observed outcomes, and we do not see strong signs of model misspecification.

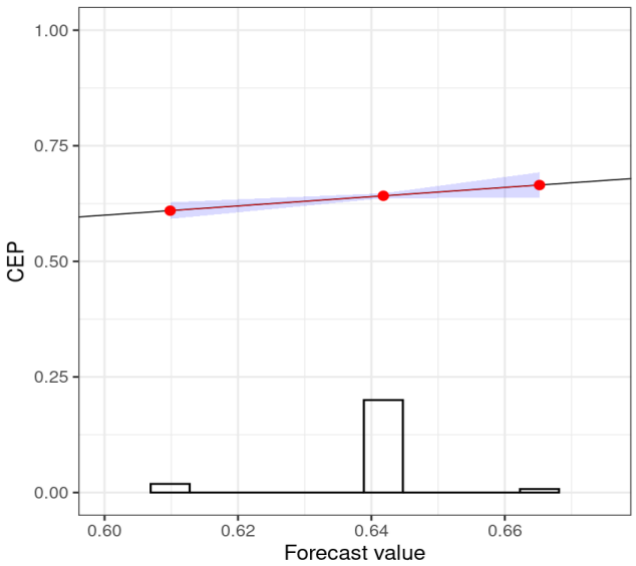

Figure 11: Reliability diagram for the non-hierarchical model.

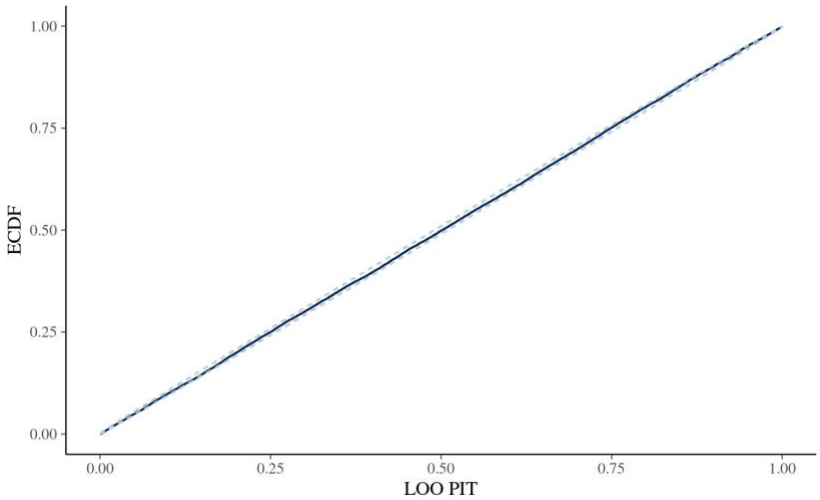

Figure 12: Loo-Pit plot for the non-hierarchical model.

Figure 13 shows the posterior predictive distributions of the serve win probability for the three point types: any point, break points, and tiebreak points. The distribution of tiebreak points seems to have a higher mean win probability with larger variance than any point. Break points are centred at a significantly lower win probability, confirming that the model captures a reduced chance of winning on break points. This is in line with the estimates in Table 1.

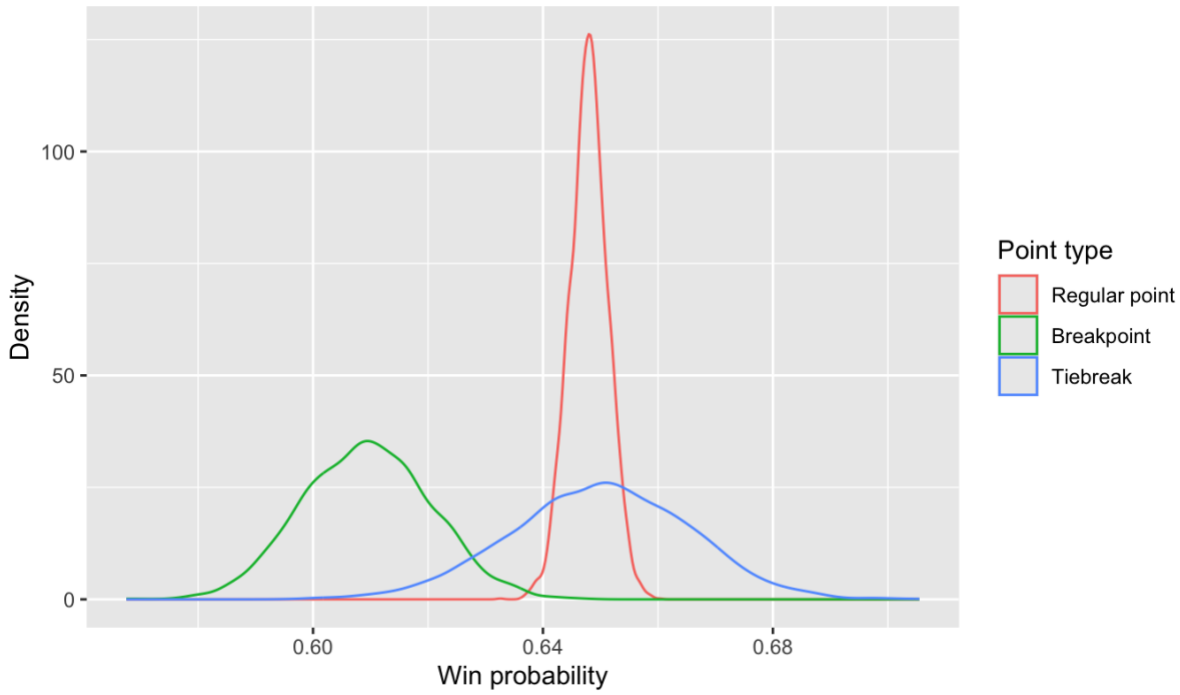

Figure 13: Posterior predictive density distribution plot for the non-hierarchical model.

Overall, the posterior predictive checks suggest that the first model provides a well-calibrated description of the data at the population level.

8.2. Posterior predictive checks for hierarchical model

The results of the PSIS_LOO diagnostics for the hierarchical model can be seen in Table 4. The elpd_loo value is utilized for comparison between the models in Section 11. P_loo gives a sense of the complexity of the model, suggesting 63.6 as the effective number of parameters. This can be explained by the hierarchical structure of the model with player-specific parameters, and the value does not seem to indicate overfitting. The looic of 31632.9 implies similarly to the first model that the second model is reasonably greedy and does not appear to overfit the data.

Measure	**Estimate**	**SE**
elpd_loo	- 15818.6	34.2
p_loo	63.6	0.7
looic	31632.9	87.9

Table 4: PSIS-Loo diagnostics of the hierarchical model.

A reliability diagram, seen in Figure 14, was further utilized to evaluate the fit of the second model. As the red step line lines mostly quite close to the perfect calibration black line, indicating that the predictions of the model somewhat match the observed frequencies, particularly for the values 0.60-0.68. Below this interval the model predicts a slightly higher win probability, whereas above this interval the predictions are slightly lower. This can be explained by the histogram, indicating fewer predictions in these regions. This further seems

reasonable, as few players are likely to have a serving win probability below 0.60 or above 0.70.

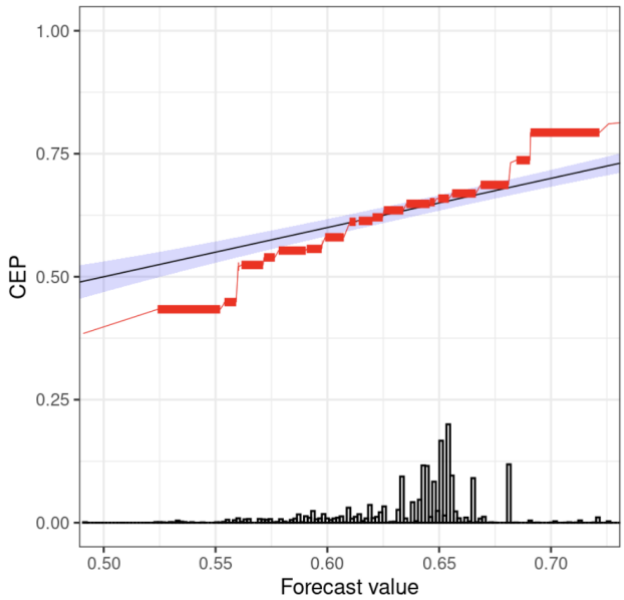

Figure 14: Reliability diagram for the hierarchical model.

The Loo-Pit plot, displayed in Figure 15, was further used to evaluate the model. This also indicates a well-calibrated model, as the Loo-Pit values are distributed in line with the diagonal and no clear deviations are detected.

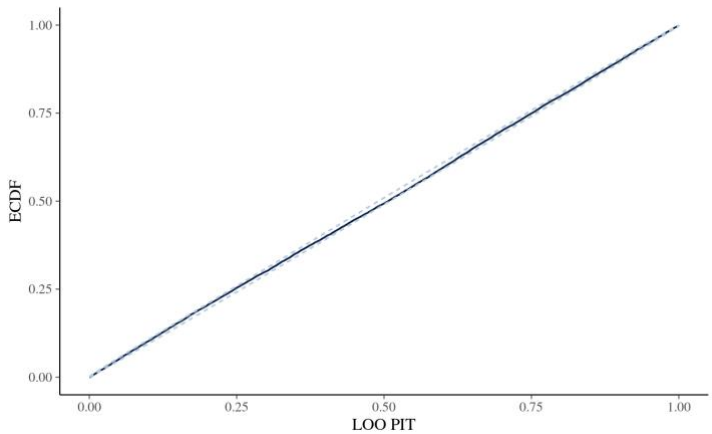

Figure 15: Loo-Pit plot for the hierarchical model.

Finally, Figure 16 shows the posterior predictive distributions of the server win probability for the three point types in the hierarchical model: regular points, breakpoints, and tiebreak points. In line with the first model, the second model implies a small but consistent decrease in win probability on break points and large variance among tiebreak points. Although the data seems to imply that many players have a higher win probability on tie breaks, this must be regarded with a degree of scepticism due to the high variance of tiebreak win probability caused by the low proportion of such data points. The hierarchical model predicts that 87% of players experience a decrease in win probability on breakpoints, with the change on tiebreak points being inconclusive.

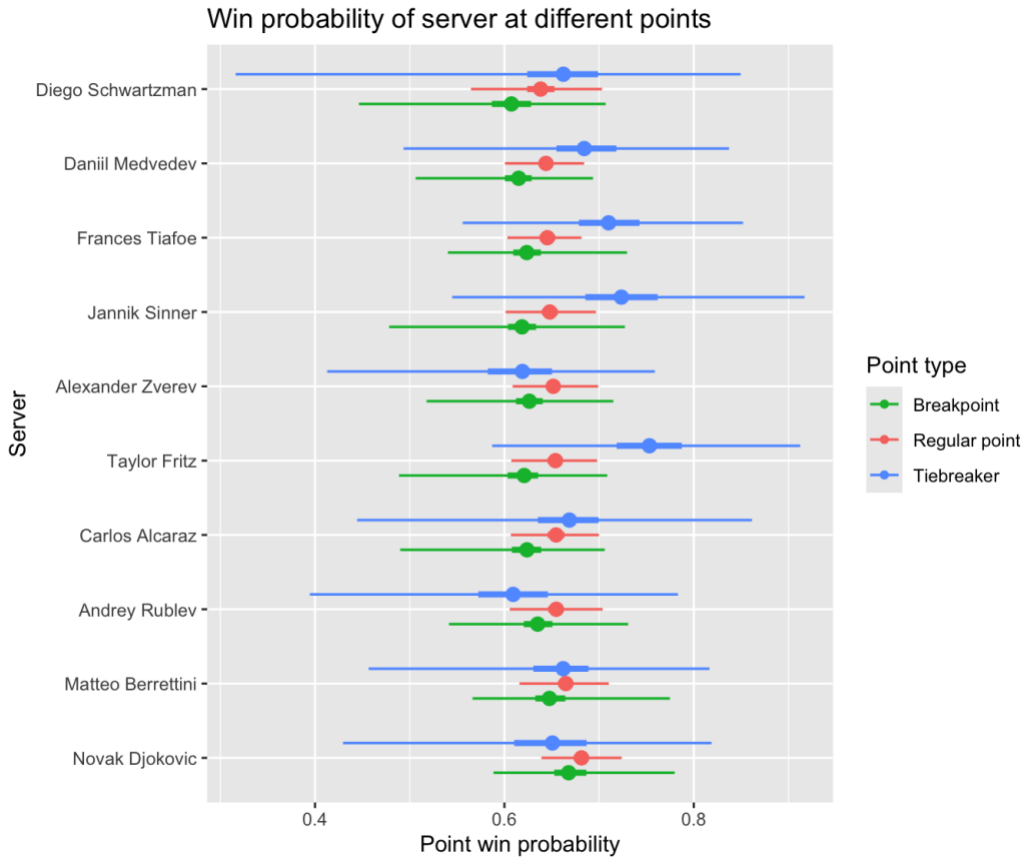

Figure 16: Posterior predictive medians and 5% and 25% top and bottom quantiles of win probability at different point types for the top 10 players in the dataset.

9. Predictive performance assessment

9.1. Non-hierarchical model

We included a ROC curve with an AUC calculation for the first model, as seen in Figure 17.

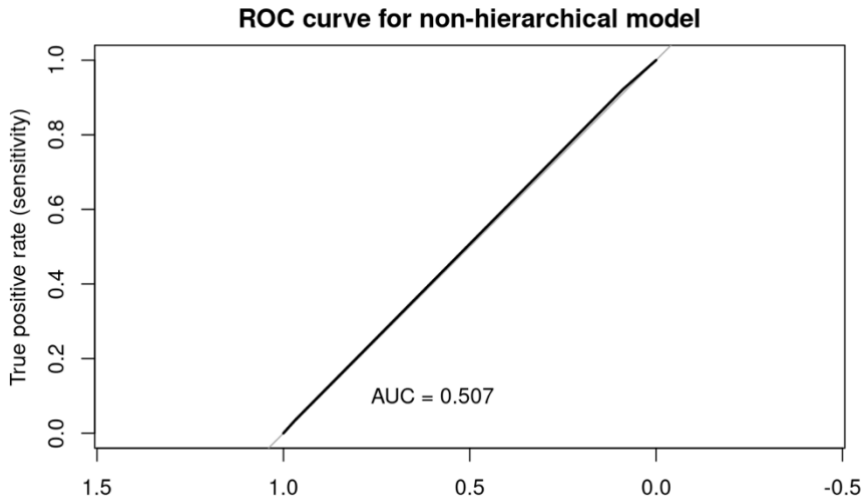

Figure 17: ROC curve for the non-hierarchical model.

We see that the model is almost no better than random at classifying won or lost points. However, overall serve point win classification is not the objective of this study. On the

contrary, we are looking at the difference pressure makes on specific points, making the ROC and AUC calculations uninformative for this study.

9.2. Hierarchical model

A ROC curve with an AUC calculation was also created for the second model, displayed in Figure 18. Based on the ROC curve it can be concluded that the model slightly detects the effect of pressure but still is not significantly better than random guessing.

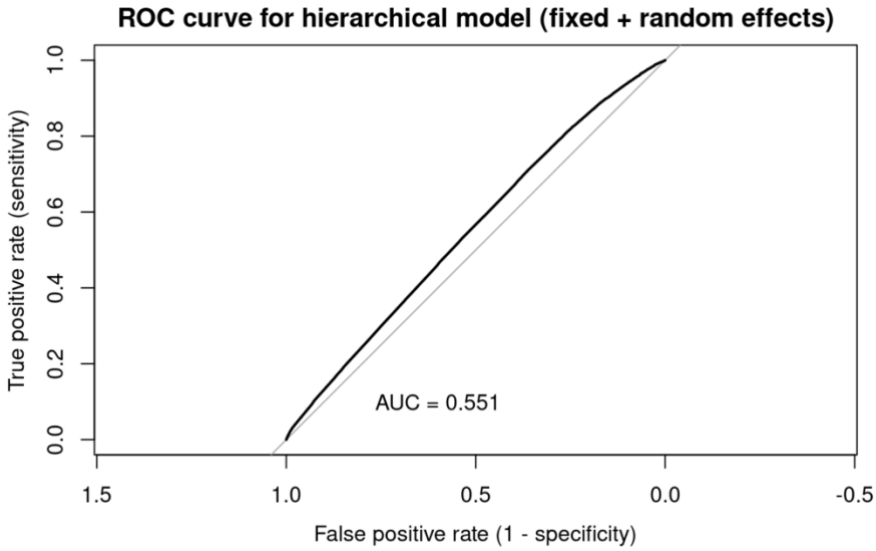

Figure 18: ROC curve for the hierarchical model.

10. Sensitivity analysis

10.1. Non-hierarchical model

The sensitivity analysis for the non-hierarchical model was conducted by creating three alternative selections of priors, as seen in Table 5. The three additional with wide, narrow, and informative priors showed that the model is stable to variations in priors, as similar results were produced for each scenario.

Param.	Prior	Est.	Err.
Intercept	$\mathcal{N}(0,2^2)$	0.58	0.01
BP	$\mathcal{N}(0,1^2)$	-0.14	0.05
TB	$\mathcal{N}(0,1^2)$	0.10	0.08

Original priors

Param.	Prior	Est.	Err.
Intercept	$\mathcal{N}(0,1^2)$	0.58	0.01
BP	$\mathcal{N}(0,0.5^2)$	-0.13	0.05
TB	$\mathcal{N}(0,0.5^2)$	0.10	0.07

Narrow priors

Param.	Prior	Est.	Err.
Intercept	$\mathcal{N}(0,5^2)$	0.58	0.01
BP	$\mathcal{N}(0,2^2)$	-0.14	0.05
TB	$\mathcal{N}(0,1^2)$	0.10	0.08

Wide priors

Param.	Prior	Est.	Err.
Intercept	$\mathcal{N}(0,5^2)$	0.58	0.01
BP	$\mathcal{N}(0,2^2)$	-0.13	0.05
TB	$\mathcal{N}(0,1^2)$	0.08	0.07

Informative priors

Table 5: Sensitivity analysis table for non-hierarchical model.

In Figure 19, the posterior densities of each scenario for each parameter are visualized. The densities are very similar, with the informative prior exhibiting the lowest variance, as expected. These results imply that the non-hierarchical model is unsensitive to the selection of prior distributions. Thus, our original choice of priors for the model are satisfactory.

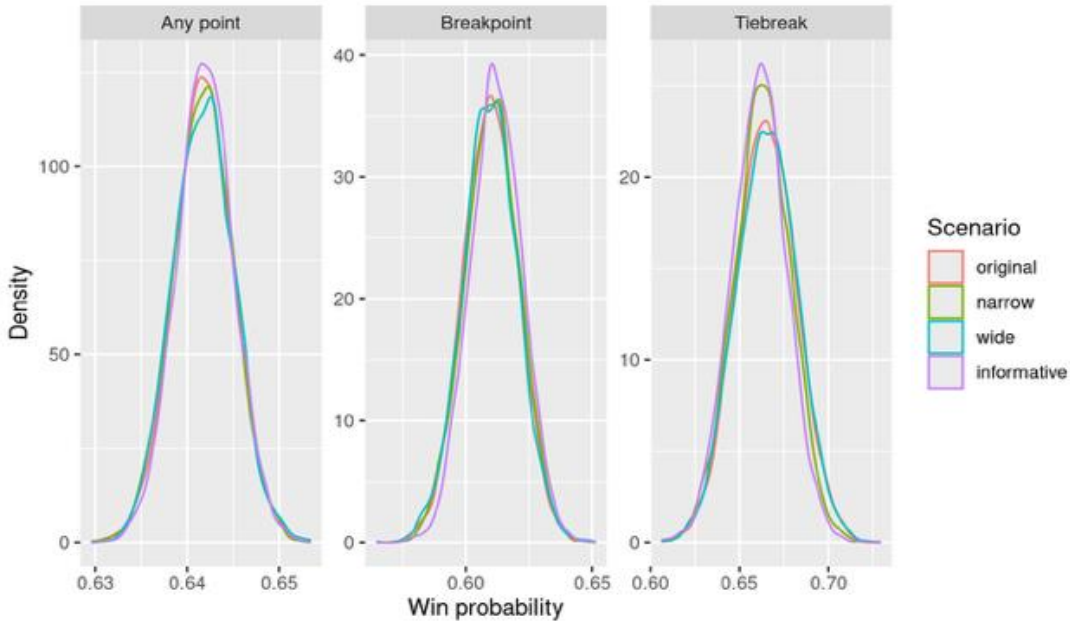

Figure 19: Posterior densities for each parameter and prior selection.

10.2. Hierarchical model

To analyse the prior sensitivity of the hierarchical model, we chose two new sets of priors on which to run the MCMC simulation, outlined in Table 6. The alternative priors were chosen to be narrow and wide, with the following interpretation.

Narrow: Average win probability prior has lower variance, and both population level and group level breakpoint and tiebreak effects have high mass near 0.

Wide: Average win probability, and population and group level breakpoint and tiebreak effects have weakly informative priors.

We saw convergence for all our key estimates.

Param.	**Prior**	**Est.**	**Err.**
α	$\mathcal{N}(0,2^2)$	0.49	0.03
β_{bp}	$\mathcal{N}(0,1^2)$	-0.12	0.05
β_{tb}	$\mathcal{N}(0,1^2)$	0.09	0.12
u_0	$t_3(0,2.5)$	0.20	0.03
u_{bp}	$t_3(0,2.5)$	0.09	0.06
u_{tb}	$t_3(0,2.5)$	0.35	0.14

Original priors

Param.	**Prior**	**Est.**	**Err.**
α	$\mathcal{N}(0,2)$	0.49	0.03
β_{bp}	$\mathcal{N}(0,0.5)$	-0.12	0.05
β_{tb}	$\mathcal{N}(0,0.5)$	0.09	0.12
u_0	$t_3(0,2.5)$	0.20	0.03
u_{bp}	$\text{Exp}(1)$	0.09	0.06
u_{tb}	$\text{Exp}(1)$	0.34	0.14

Narrow priors

Param.	**Prior**	**Est.**	**Err.**
α	$\mathcal{N}(0,10)$	0.49	0.03
β_{bp}	$\mathcal{N}(0,5)$	-0.12	0.05
β_{tb}	$\mathcal{N}(0,5)$	0.10	0.13
u_0	$t_3(0,5)$	0.20	0.03
u_{bp}	$t_3(0,5)$	0.09	0.07
u_{tb}	$t_3(0,5)$	0.36	0.14

Wide priors

Table 6: Priors for sensitivity analysis of the hierarchical model.

11. Model comparison

To compare the two models, we used PSIS-LOO-CV as recommended in the course. The resulting expected log predictive density ($\mathrm{elpd}_{\mathrm{loo}}$) are indicated in Table 3 and Table 4 in section 8, and a comparison can be seen in Table 7 below. The non-hierarchical model has an elpd difference of -36.8 and a standard error difference of 8.9 compared to the hierarchical baseline. The difference is greater than four standard errors, which can be interpreted as strong evidence in favor of the hierarchical model. The hierarchical structure appears to capture important heterogeneity between servers that the non-hierarchical model cannot explain.

Model	**Elpd difference**	**Standard error difference**
Hierarchical model	0.0	0.0
Non-hierarchical model	-36.8	8.9

Table 7: elpd and standard error differences between the two models.

The PSIS diagnostics did not indicate any problematic observations for either model, as seen in Figure 20 below. No $\hat{k}$ values exceeding 0.5 were detected, so we can be confident in the PSIS-LOO-CV comparison. Overall, we select the hierarchical model as our main model for inference of pressure effects.

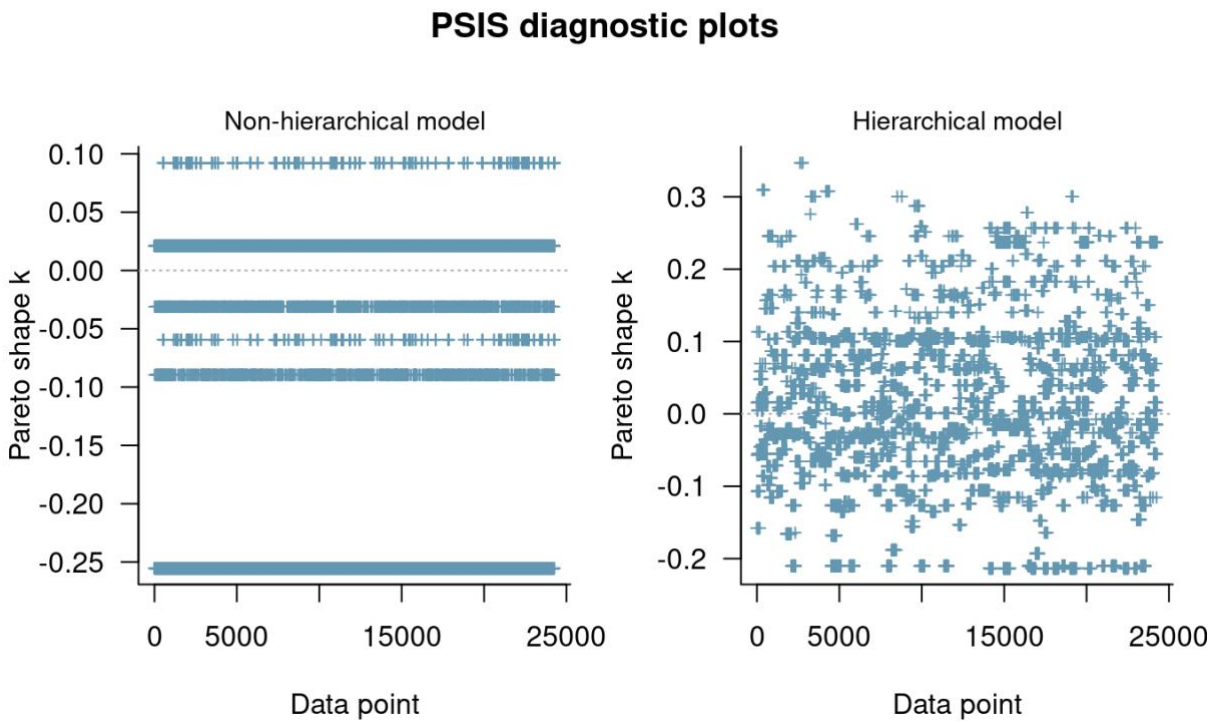

Figure 20: PSIS diagnostic plots for each model.

12. Discussion of issues and potential improvements

Potential problems and improvements need to be considered for the models. First, the predictive performance could be improved for both models as the ROC AUC are only slightly better than random guessing. This could be improved by including e.g. the following parameters to more accurately model pressure: set point, match in tournament and who the returning player is. The priors could also be more detailed, for example as random effects of

the servers come from a single multivariate normal distribution. In reality players might be more heterogenous. In addition, the report could also have been expanded with additional models beyond the two used.

13. Conclusion

The goal of this project was to investigate how pressure situations affect server pointwise win probabilities in professional tennis. We used a sample of 104 randomly selected Grand Slam matches, with approximately 24 000 individual points. We focused on two types of high-pressure serving situations: break points against the server and points played in tiebreaks. We compared these situations to regular points using Bayesian logistic regression models fitted with MCMC methods.

Across both the non-hierarchical and hierarchical models, the main conclusion is consistent. The server has a lower point win probability on break points compared to regular points, with the posterior for break point coefficient being concentrated below zero and the posterior predictive distributions showing a lower win probability when facing break points. However, the pressure effect of serving during tiebreak points is estimated to be small and have a lot of variance, with some players experiencing an increase in win probability, and other players showing a decrease in win probability when serving on tiebreak points. Thus, we cannot confidently claim a systematic advantage or disadvantage in tiebreaks at the population level.

Posterior predictive checks, reliability diagrams, and LOO PIT diagnostics indicate that models provide a reasonably well calibrated description of the observed data. The ROC and AUC analyses show that neither model is especially useful for point level classification, however, this is in line with our expectations as the goal of the study was to explore pressure effects rather than to build a high accuracy classifier. The sensitivity analyses further show that the results are stable to changes in priors, suggesting that the conclusions are mainly driven by the data.

Model comparison with LOO-CV clearly favors the hierarchical specification, which accounts for player-specific variation in baseline serve point win probability and pressure effects. This indicates that heterogeneity between players is important for accurately capturing serve performance under pressure, even though population level results about pressure remain similar. Overall, our analysis supports the view that pressure at break point measurably reduces server point win probability, while tiebreak points primarily increase variance rather than systematically improving or reducing point win serve point win probability.

14. Self-reflection

While making the project and report our group learned to apply the Bayesian workflow to a real-world problem. We learned to choose suitable data, e.g. not choosing a too big dataset, pick suitable models based on the data and problem as well as to evaluate and interpret the outputs. A key learning in this specific project was the benefits of using a hierarchical model to get more accurate results. We also learned to utilize PSIS-LOO metrics, reliability diagrams, LOO-PIT diagnostics and posterior predictive distribution visualizations as part of assessing the posterior predictive performance. Finally, we learned to utilize BRMS more efficiently in specifying models. Beyond the technical learnings our group additionally learned to cooperate and coordinate a project. We learned the importance of communication

in order to keep each group member up to date as well as the benefits of holding in-person meetings regularly to discuss progress and solve challenges together. As the project contained quite a few challenges and setbacks, we learned to work together as a group to solve these.

15. References

Gelman, A., Carlin, J.B., Stern, H.S., Dunson, D.B., Vehtari, A. and Rubin, D.B. (2021). *Bayesian Data Analysis*. 3rd ed. CRC Press.

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